

ProbOS — Overall Plan

Nisong Monyimba

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Purpose

This document is the single source of truth for the full ProbOS roadmap. Earlier planning used loose “Year 1 / Year 2 / Year 3” labels that did not line up with actual calendar years (Year 1 covered 4 months, Year 2 covered 8 months). This document replaces those labels with a consistent Month 1 through Month 24 numbering. No month number here is retroactively changed once work on it begins — if a month’s plan turns out to be wrong once we reach it, we correct it in that month’s own `main.tex` (as already done for Month 1 and Month 2), not here.

Total planned horizon

24 months (2 years). Months 1–12 are concretely planned at weekly granularity. Months 13–24 are directional and will be replanned in detail as each one approaches, the same way Month 2 was just replanned at the end of Month 1.

Status legend

- **DONE** — complete, documented in its own monthly plan PDF under `docs/monthly_plans/`
- **PLANNED** — detailed week-by-week plan exists
- **DIRECTIONAL** — theme and goal known, not yet week-by-week planned

Months 1–4 — Kernel Foundation

Month	Theme	Key deliverables	Status
1	Distribution + Model ABCs, Monte Carlo kernel, PDSL v0.1	Distribution/Model ABCs, BatteryModel2Cell, MonteCarloEngine, SobolSensitivity, ProvenanceTracker, C++/OpenMP kernel (7x), PDSL compiler v0.1, 3 cross-discipline examples, 264 Python + 13 C++ tests	DONE
2	Sequential inference + C++ into Python package	ParticleFilter (validated vs Kalman ground truth), pybind11 bindings, FastAPI service layer, cross-discipline filtering examples	PLANNED

Month	Theme	Key deliverables	Status
3	GPU kernel path, pharma domain, automotive positioning, PDSL v0.2, comprehensive C++/kernel sweep	GPU kernel path (honestly benchmarked), <code>PharmaStabilityModel</code> (validated against Gonzalez-Gonzalez et al. 2023), automotive/EV positioning materials, PDSL v0.2 (control flow), C++/Python parameter-spread discrepancy fixed at the root, 408 Python + 5 C++ test executables	DONE
3.5	Injection molding process qualification (Week 14, research-gate first)	<code>InjectionMoldingProcessModel</code> , reusing the existing engine for a third domain; real validation contingent on a genuine Day 1 research gate; fourth live demo tab	PLANNED
4	Dashboard architecture and core build	Framework decision (Streamlit vs. React), core dashboard for the validated domains, P05/P50/P95 visualization, Sobol chart, error handling and a real cold-start user-flow test	PLANNED

Months 5–12 — Directional (formerly “Year 2”)

Month	Directional theme
5–8	Consolidation of Months 1–4 capabilities into a stable, documented, installable package. Expanded example library across more domains. Performance work informed by real profiling from earlier months, not assumed in advance.
9–12	Broader platform concerns: packaging for external users, documentation, and any capabilities that Months 1–8 reveal are actually needed (rather than pre-specified now).

Honesty note: the specific technical content of Months 5–12 was previously drafted in detail before Month 1 was even finished. That level of pre-specification was premature — Month 1 alone surfaced priorities (e.g. the particle filter becoming Month 2’s centrepiece) that were not obvious in advance. Months 5–12 are intentionally left as themes here and will be planned in full, in their own `docs/monthly_plans/monthN/main.tex`, one month at a time as we approach them.

Months 13–24 — Directional (formerly “Year 3”)

Month	Directional theme
13–18	Scale and robustness of whatever platform shape has emerged from Months 1–12.

Month	Directional theme
19–24	Maturity, external usage, and closing out the 24-month horizon with a clear-eyed assessment of what should come next (a new 24-month plan, or a narrower maintenance phase).

Process going forward

1. This document (`docs/monthly_plans/overall/main.tex`) is updated whenever the total horizon or month numbering changes — not on every month’s completion.
2. Each individual month gets its own `docs/monthly_plans/monthN/main.tex` plus compiled PDF, written and compiled *before* that month’s work begins.
3. Each month closes with a retrospective appended to (or replacing the plan content of) that same month’s PDF, as already done for Month 1.
4. `check_ci.sh` is run after every push; mypy strict and ruff stay clean; no benchmark claim is made without a genuine measurement backing it.